

Lista de atividades de Análise de Sobrevida

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Questão 2

Enunciado

Implemente os modelos exponencial, gamma, log-normal e log-logístico no stan e ajuste os modelos implementados ao banco de dados sobrevivencia::isoladores.

Carregando pacotes

```
library(rstan)
```

Warning: pacote 'rstan' foi compilado no R versão 4.5.3

Carregando pacotes exigidos: StanHeaders

Warning: pacote 'StanHeaders' foi compilado no R versão 4.5.3

rstan version 2.32.7 (Stan version 2.32.2)

For execution on a local, multicore CPU with excess RAM we recommend calling `options(mc.cores = parallel::detectCores())`.

To avoid recompilation of unchanged Stan programs, we recommend calling `rstan_options(auto_write = TRUE)`

For within-chain threading using ``reduce_sum()`` or ``map_rect()`` Stan functions, change ``threads_per_chain`` option:

```
rstan_options(threads_per_chain = 1)
```

Do not specify `'-march=native'` in `'LOCAL_CPPFLAGS'` or a `Makevars` file

```
library(sobrevivencia)
```

Carregando pacotes exigidos: survival

```
library(survival)

rstan_options(auto_write = TRUE)
options(mc.cores = parallel::detectCores())
```

Banco de dados

```
data(isoladores, package = "sobrevivencia")

head(isoladores)
```

	tempo	evento
1	151	1
2	164	1
3	336	1
4	365	1
5	403	1
6	454	1

Preparação dos dados

```
N <- nrow(isoladores)

stan_data <- list(
  N = N,
  t = isoladores$tempo,
  d = isoladores$status
)
```

Modelo Exponencial

Assumimos:

$$T_i \sim \text{Exp}(\lambda)$$

A função de sobrevivência é dada por:

$$S(t) = e^{-\lambda t}$$

Código Stan

```
exp_stan <- '  
data {  
  int<lower=1> N;  
  vector<lower=0>[N] t;  
  array[N] int<lower=0,upper=1> d;  
}  
  
parameters {  
  real<lower=0> lambda;  
}  
  
model {  
  
  lambda ~ gamma(0.01, 0.01);  
  
  for (i in 1:N) {  
  
    if (d[i] == 1)  
      target += exponential_lpdf(t[i] | lambda);  
  
    else  
      target += exponential_lccdf(t[i] | lambda);  
  
  }  
}  
'
```

Ajuste do modelo

```
mod_exp <- stan_model(model_code = exp_stan)
```

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.2 from <https://cran.r-project.org/bin/windows/Rtools/>.

Trying to compile a simple C file

```
Running "C:/PROGRA~1/R/R-45~1.2/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 14.3.0'
gcc -I"C:/PROGRA~1/R/R-45~1.2/include" -DNDEBUG -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/Rcpp/include/" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/StanHeaders/include/" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/Laboratorio/AppData/Local/R/win-library/4.5/rstan/include" -
DEIGEN_NO_DEBUG -DBOOST_DISABLE_ASSERTS -DBOOST_PENDING_INTEGER_LOG2_HPP -
DSTAN_THREADS -DUSE_STANC3 -DSTRICT_R_HEADERS -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION -
D_HAS_AUTO_PTR_ETC=0 -include "C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp" -
std=c++1y -I"C:/rtools45/x86_64-w64-mingw32.static.posix/include" -
O2 -Wall -std=gnu2x -mfpmath=sse -msse2 -mstackrealign -
c foo.c -o foo.o
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
In file included from C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
      from C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
      from C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
      from <command-line>:
C:/Users/Laboratorio/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/src/Core/util/M
679 | #include <cmath>
      |          ~~~~~~
compilation terminated.
make: *** [C:/PROGRA~1/R/R-45~1.2/etc/x64/Makeconf:289: foo.o] Error 1
```

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.2 from
<https://cran.r-project.org/bin/windows/Rtools/>.

```
fit_exp <- sampling(
  mod_exp,
  data = stan_data,
  iter = 4000,
  warmup = 2000,
  chains = 4,
```

```
seed = 123
)
```

Error in eval(expr, envir) :

Exception: variable does not exist; processing stage=data initialization; variable name=d,

failed to create the sampler; sampling not done

Resumo

```
print(fit_exp)
```

Stan model 'anon_model' does not contain samples.

Modelo Gamma

Assumimos:

$$T_i \sim \text{Gamma}(\alpha, \beta)$$

Código Stan

```
gamma_stan <- '  
data {  
  int<lower=1> N;  
  vector<lower=0>[N] t;  
  array[N] int<lower=0,upper=1> d;  
}  
  
parameters {  
  real<lower=0> alpha;  
  real<lower=0> beta;  
}  
  
model {  
  
  alpha ~ gamma(0.01,0.01);  
  beta ~ gamma(0.01,0.01);  
}
```

```

for (i in 1:N) {

  if (d[i] == 1)
    target += gamma_lpdf(t[i] | alpha, beta);

  else
    target += gamma_lccdf(t[i] | alpha, beta);

}

}
'

```

Ajuste do modelo

```
mod_gamma <- stan_model(model_code = gamma_stan)
```

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Trying to compile a simple C file

```

Running "C:/PROGRA~1/R/R-45~1.2/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 14.3.0'
gcc -I"C:/PROGRA~1/R/R-45~1.2/include" -DNDEBUG -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/Rcpp/include/" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/StanHeaders/include/" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/Laboratorio/AppData/Local/R/win-library/4.5/rstan/include" -
DEIGEN_NO_DEBUG -DBOOST_DISABLE_ASSERTS -DBOOST_PENDING_INTEGER_LOG2_HPP -
DSTAN_THREADS -DUSE_STANC3 -DSTRICT_R_HEADERS -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION -
D_HAS_AUTO_PTR_ETC=0 -include "C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp" -
std=c++1y -I"C:/rtools45/x86_64-w64-mingw32.static.posix/include" -
O2 -Wall -std=gnu2x -mfpmath=sse -msse2 -mstackrealign -
c foo.c -o foo.o

```

```
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
In file included from C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
      from C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
      from C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
      from <command-line>:
C:/Users/Laboratorio/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/src/Core/util/
679 | #include <cmath>
      |          ~~~~~~
compilation terminated.
make: *** [C:/PROGRA~1/R/R-45~1.2/etc/x64/Makeconf:289: foo.o] Error 1
```

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.2 from
<https://cran.r-project.org/bin/windows/Rtools/>.

```
fit_gamma <- sampling(
  mod_gamma,
  data = stan_data,
  iter = 4000,
  warmup = 2000,
  chains = 4,
  seed = 123
)
```

```
Error in eval(expr, envir) :
  Exception: variable does not exist; processing stage=data initialization; variable name=d
```

failed to create the sampler; sampling not done

Resumo

```
print(fit_gamma)
```

Stan model 'anon_model' does not contain samples.

Modelo Log-Normal

Assumimos:

$$\log(T_i) \sim N(\mu, \sigma^2)$$

Código Stan

```
lnorm_stan <- '  
data {  
  int<lower=1> N;  
  vector<lower=0>[N] t;  
  array[N] int<lower=0,upper=1> d;  
}  
  
parameters {  
  real mu;  
  real<lower=0> sigma;  
}  
  
model {  
  
  mu ~ normal(0,100);  
  sigma ~ cauchy(0,5);  
  
  for (i in 1:N) {  
  
    if (d[i] == 1)  
      target += lognormal_lpdf(t[i] | mu, sigma);  
  
    else  
      target += lognormal_lccdf(t[i] | mu, sigma);  
  
  }  
}  
'
```

Ajuste do modelo

```
mod_lnorm <- stan_model(model_code = lnorm_stan)
```

WARNING: Rtools is required to build R packages, but is not currently installed.

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Trying to compile a simple C file

Running "C:/PROGRA~1/R/R-45~1.2/bin/x64/Rcmd.exe" SHLIB foo.c

using C compiler: 'gcc.exe (GCC) 14.3.0'

```
gcc -I"C:/PROGRA~1/R/R-45~1.2/include" -DNDEBUG -I"C:/Users/Laboratorio/AppData/Local/R/win-  
library/4.5/Rcpp/include/" -I"C:/Users/Laboratorio/AppData/Local/R/win-  
library/4.5/RcppEigen/include/" -I"C:/Users/Laboratorio/AppData/Local/R/win-  
library/4.5/RcppEigen/include/unsupported" -I"C:/Users/Laboratorio/AppData/Local/R/win-  
library/4.5/BH/include" -I"C:/Users/Laboratorio/AppData/Local/R/win-  
library/4.5/StanHeaders/include/src/" -I"C:/Users/Laboratorio/AppData/Local/R/win-  
library/4.5/StanHeaders/include/" -I"C:/Users/Laboratorio/AppData/Local/R/win-  
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -  
I"C:/Users/Laboratorio/AppData/Local/R/win-library/4.5/rstan/include" -  
DEIGEN_NO_DEBUG -DBOOST_DISABLE_ASSERTS -DBOOST_PENDING_INTEGER_LOG2_HPP -  
DSTAN_THREADS -DUSE_STANC3 -DSTRICT_R_HEADERS -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION -  
D_HAS_AUTO_PTR_ETC=0 -include "C:/Users/Laboratorio/AppData/Local/R/win-  
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp" -  
std=c++1y -I"C:/rtools45/x86_64-w64-mingw32.static.posix/include" -  
O2 -Wall -std=gnu2x -mfpmath=sse -msse2 -mstackrealign -  
c foo.c -o foo.o
```

cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C

In file included from C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,

from C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,

from C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,

from <command-line>:

C:/Users/Laboratorio/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/src/Core/util/

```
679 | #include <cmath>
```

```
| ~~~~~
```

compilation terminated.

make: *** [C:/PROGRA~1/R/R-45~1.2/etc/x64/Makeconf:289: foo.o] Error 1

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.2 from <https://cran.r-project.org/bin/windows/Rtools/>.

```
fit_lnorm <- sampling(
  mod_lnorm,
  data = stan_data,
  iter = 4000,
  warmup = 2000,
  chains = 4,
  seed = 123
)
```

Error in eval(expr, envir) :

Exception: variable does not exist; processing stage=data initialization; variable name=d

failed to create the sampler; sampling not done

Resumo

```
print(fit_lnorm)
```

Stan model 'anon_model' does not contain samples.

Modelo Log-Logístico

A função de sobrevivência do modelo log-logístico é:

$$S(t) = \frac{1}{1 + (t/\alpha)^\beta}$$

Neste exercício implementamos manualmente a densidade e a função de sobrevivência do modelo log-logístico utilizando funções definidas pelo usuário no Stan.

Código Stan

```
llogis_stan <- '
functions {
  real my_loglogistic_lpdf(real t, real alpha, real beta) {
```

```

    return log(beta) - log(alpha) +
           (beta - 1) * (log(t) - log(alpha)) -
           2 * log1p(pow(t / alpha, beta));

}

real my_loglogistic_lccdf(real t, real alpha, real beta) {

    return -log1p(pow(t / alpha, beta));

}

}

data {

    int<lower=1> N;
    vector<lower=0>[N] t;
    array[N] int<lower=0,upper=1> d;

}

parameters {

    real<lower=0> alpha;
    real<lower=0> beta;

}

model {

    alpha ~ gamma(0.01,0.01);
    beta ~ gamma(0.01,0.01);

    for (i in 1:N) {

        if (d[i] == 1)
            target += my_loglogistic_lpdf(t[i] | alpha, beta);

        else
            target += my_loglogistic_lccdf(t[i] | alpha, beta);

    }

}

```

Ajuste do modelo

```
mod_llogis <- stan_model(model_code = llogis_stan)
```

WARNING: Rtools is required to build R packages, but is not currently installed.

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Trying to compile a simple C file

```
Running "C:/PROGRA~1/R/R-45~1.2/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 14.3.0'
gcc -I"C:/PROGRA~1/R/R-45~1.2/include" -DNDEBUG -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/Rcpp/include/" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/StanHeaders/include/" -I"C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/Laboratorio/AppData/Local/R/win-library/4.5/rstan/include" -
DEIGEN_NO_DEBUG -DBOOST_DISABLE_ASSERTS -DBOOST_PENDING_INTEGER_LOG2_HPP -
DSTAN_THREADS -DUSE_STANC3 -DSTRICT_R_HEADERS -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION -
D_HAS_AUTO_PTR_ETC=0 -include "C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp" -
std=c++1y -I"C:/rtools45/x86_64-w64-mingw32.static.posix/include" -
O2 -Wall -std=gnu2x -mfpmath=sse -msse2 -mstackrealign -
c foo.c -o foo.o
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
In file included from C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
               from C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
               from C:/Users/Laboratorio/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
               from <command-line>:
C:/Users/Laboratorio/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/src/Core/util/M
679 | #include <cmath>
    |          ~~~~~~
```

compilation terminated.

make: *** [C:/PROGRA~1/R/R-45~1.2/etc/x64/Makeconf:289: foo.o] Error 1

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.2 from <https://cran.r-project.org/bin/windows/Rtools/>.

```
fit_llogis <- sampling(  
  mod_llogis,  
  data = stan_data,  
  iter = 4000,  
  warmup = 2000,  
  chains = 4,  
  seed = 123  
)
```

Error in eval(expr, envir) :

Exception: variable does not exist; processing stage=data initialization; variable name=d

failed to create the sampler; sampling not done

Resumo

```
print(fit_llogis)
```

Stan model 'anon_model' does not contain samples.

Observações

Para observações censuradas utilizamos as funções:

- *_lccdf()

pois:

$$P(T > t) = S(t)$$

Além disso, em Stan:

```
x ~ dist(...)
```

é equivalente a:

```
target += dist_lpdf(...)
```